



BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) - 2024
SEMESTER –V

Course Code: BCA57205

Course Name: Machine Learning

Weekly Contact Hours: 3L+2P

Credits: 4

Total Allotted Hours: 45L + 30P

Course Objective: The main aim of this course is to apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.

Pre-requisite(s): Knowledge of AI and algorithm.

Course Outcome (COs): After the completion of the course, students would be able to:

CO1: Define Machine Learning and differentiate it with Artificial Intelligence.

CO2: Explain and analyze the significance of using supervised, unsupervised and reinforcement learning.

CO3: Organize and evaluate data preprocessing and training and testing model.

CO4: Relate AI with Machine Learning to solve various business problems.

CO5: Demonstrate, evaluate and develop different Ensemble learning algorithms

Module 1: Introduction to Machine Learning

[8H(L)]

Understanding machine learning -principle of machine learning, compare machine learning with Artificial Intelligence, defining big data- big data in context with machine learning, leveraging the power of machine learning, descriptive analytics - predictive analytics.

Practical:

[4H(P)]

Implementation of data visualization and representation using python libraries.

Module 2: Types of Learning and its applications

[12H(L)]

The roles of statistics and data mining with machine learning, approaches to machine learning, supervised learning, unsupervised learning, reinforcement learning, neural networks. Applying machine learning - understanding machine learning techniques, tying machine learning methods to outcomes, applying machine learning to business needs.

Practical:

[12H(P)]

Implementation of data preprocessing and machine learning models.

Module 3: Training of Model in machine learning

[10H(L)]

Looking inside machine learning, the role of algorithms, types of machine learning algorithms, training machine learning systems, data preparation, identify relevant data, the machine learning cycle.

Practical:

[4H(P)]

Implementation of training algorithms in machine learning model.



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Module 4: AI and Machine learning with Business Problems

[10H(L)]

Understanding how machine learning can help, focus on the business problem, executing a pilot project, determining the best learning model, machine learning skills. Machine learning to provide solutions to business problems, applying machine learning to patient health, proactively responding to IT Issues, protecting against fraud.

Practical:

[6H(P)]

Implementation of model training and prediction algorithms.

Module 5: Ensemble Learning

[5H(L)]

Ensemble learning model combination schemes, voting, error-correcting output codes, bagging: random forest trees, boosting: adaboost, stacking.

Practical:

[4H(P)]

Implementation of ensemble models and evaluations.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1, K4
2.	CO2	M2	K2, K4, K5
3.	CO3	M3	K3, K5
4.	CO4	M4	K2, K6
5.	CO5	M5	K2, K5, K6

Text books:

1. Introduction to Machine Learning Second Edition, The MIT Press Cambridge, Massachusetts, London, England, EthemAlpaydm.

Reference books

1. Hands-On Machine Learning with Scikit-Learn &TensorFlow, O'Reilly Media, AurélienGéron, Inc., 2nd Edition, 2022.
2. Machine Learning Algorithms: Handbook, Clever Fox Publishing, Aman Kharwal ,1st Edition, 2023.